

Internship Fusion Task 2:

Titanic Dataset: Data Cleaning and Exploratory Data Analysis (EDA) Report

Introduction

The Titanic dataset is a well-known dataset that contains data about the passengers aboard the ill-fated RMS Titanic, which sank on April 15, 1912. The dataset provides information about various attributes such as the passengers' age, sex, class, fare, and survival status. The objective of this analysis is to clean the data, perform exploratory data analysis (EDA), and derive meaningful insights about the factors that influence the survival of passengers.

This report documents our steps to clean the dataset and the insights gained through visualizations and statistical analysis.

Step-by-Step Process

Step 1: Importing Necessary Libraries

To perform data manipulation, visualization, and statistical analysis, we imported the following Python libraries:

- **pandas**: For data handling and manipulation.
- **numpy**: For numerical operations.
- **matplotlib** and **seaborn**: For data visualizations.

```
python
Copy code
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Loading the Titanic Dataset

The dataset was loaded from a CSV file using `pandas`' `read_csv` function. A preview of the first few rows was displayed to ensure that the data had been correctly imported.

```
python
Copy code
titanic_data = pd.read_csv('titanic.csv')
titanic_data.head()
```

Step 3: Data Cleaning

Data cleaning is essential to ensure that the analysis is accurate and meaningful. We focused on addressing missing values, correcting data types, and dropping irrelevant features.

a. Identifying Missing Values

We first identified missing values in the dataset using `isnull().sum()`. This function helped us understand the extent of missing values in each column.

```
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titanic_data.isnull().sum()
```

The results showed missing values in the **Age**, **Embarked**, and **Cabin** columns.

b. Handling Missing Values

- **Age:** Since age is a significant factor in our analysis, we filled the missing values in the `Age` column with the median value of the available ages.

```
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titanic_data['Age'].fillna(titanic_data['Age'].median(), inplace=True)
```

- **Embarked:** The `Embarked` column contains the port of embarkation for each passenger. As only a small number of values were missing, we filled them with the most frequent value (mode) in the column.

```
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titanic_data['Embarked'].fillna(titanic_data['Embarked'].mode()[0],
inplace=True)
```

- **Cabin:** Given that a large proportion of the `Cabin` column was missing, we decided to drop this column from our analysis.

```
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titanic_data.drop('Cabin', axis=1, inplace=True)
```

c. Data Type Corrections

We also optimized the dataset by converting columns such as `Survived` and `Pclass` into categorical data types, which reduces memory usage and improves processing speed.

```
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titanic_data['Survived'] = titanic_data['Survived'].astype('category')
titanic_data['Pclass'] = titanic_data['Pclass'].astype('category')
```

Step 4: Exploratory Data Analysis (EDA)

With a clean dataset, we performed an exploratory data analysis (EDA) to gain insights into factors that influenced survival rates.

a. Survival Rate Distribution

The first visualization was a simple count plot showing the overall survival rate. The dataset was imbalanced, with more passengers not surviving the disaster.

```
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sns.countplot(data=titanic_data, x='Survived')
plt.title('Survival Count')
plt.show()
```

Conclusion: The survival rate was around 38%, with more passengers not surviving (62%).

b. Survival Rate by Gender

Next, we analyzed how survival was affected by gender. The results showed a stark difference between male and female survival rates.

```
python
Copy code
sns.countplot(data=titanic_data, x='Survived', hue='Sex')
plt.title('Survival Count by Gender')
plt.show()
```

Conclusion: Females had a much higher survival rate than males. This finding aligns with the well-known principle of “women and children first” during the evacuation of the Titanic.

c. Survival by Passenger Class (Pclass)

Passenger class (Pclass) was another key factor in survival. We visualized the survival rates for each class to better understand the distribution.

```
python
Copy code
sns.countplot(data=titanic_data, x='Survived', hue='Pclass')
plt.title('Survival Count by Passenger Class')
plt.show()
```

Conclusion: Passengers in first class had the highest survival rate, followed by second class, with third class passengers having the lowest survival rate. This suggests that socio-economic status played a significant role in survival.

d. Age Distribution vs. Survival

To understand how age affected survival, we plotted the age distribution for passengers who survived versus those who did not.

```
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plt.figure(figsize=(10, 6))
```

```
sns.histplot(titanic_data[titanic_data['Survived'] == 1]['Age'],
label='Survived', color='green', kde=True)
sns.histplot(titanic_data[titanic_data['Survived'] == 0]['Age'], label='Not
Survived', color='red', kde=True)
plt.legend()
plt.title('Age Distribution by Survival')
plt.show()
```

Conclusion: Younger passengers, especially children, had a higher likelihood of survival. Older passengers, on the other hand, were less likely to survive.

e. Fare vs. Survival

Since fare is closely tied to passenger class, we also visualized how fare impacted survival rates.

```
python
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sns.boxplot(data=titanic_data, x='Survived', y='Fare')
plt.title('Fare vs. Survival')
plt.show()
```

Conclusion: Passengers who paid higher fares had a higher survival rate, likely because they were in higher classes.

f. Correlation Heatmap

Lastly, we generated a correlation heatmap to explore relationships between the various numerical features in the dataset.

```
python
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plt.figure(figsize=(10, 8))
sns.heatmap(titanic_data.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```

Conclusion: There was a positive correlation between fare and survival, as well as a negative correlation between class and survival (lower class number means higher class, leading to higher survival rates).

Key Insights from the Analysis

1. **Gender Influence:** Gender had a strong impact on survival rates, with females being significantly more likely to survive compared to males.
2. **Class Influence:** Passenger class was another critical factor, with first-class passengers enjoying the highest survival rate and third-class passengers experiencing the lowest.
3. **Age Influence:** Age was a key determinant, with younger passengers (especially children) having a greater chance of survival.

4. **Fare:** Fare, which is linked to passenger class, showed a strong positive relationship with survival. Higher-paying passengers tended to survive at higher rates.
 5. **Combination of Factors:** Gender, class, and age combined provide the most comprehensive view of survival likelihood. Females in first class, for example, had the highest survival rates.
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Conclusion

The Titanic dataset provides valuable insights into the demographics and other factors that contributed to survival during the disaster. Through careful data cleaning and EDA, we were able to highlight the importance of gender, age, and socio-economic status in influencing survival. These findings not only help us understand historical events but also serve as a useful exercise in analyzing real-world datasets and drawing conclusions from data.

In a machine learning context, these insights could also guide the creation of predictive models to estimate survival probabilities based on a combination of demographic factors.